



UNIT 1: ECOLOGY – MCQs

1. Levels of Ecological Organization

1. The smallest unit of ecological study is:

- A. Population
- B. Ecosystem
- C. Organism
- D. Community

☒ **Answer: C**

2. A group of individuals of the same species in an area is called:

- A. Community
- B. Ecosystem
- C. Population
- D. Biosphere

☒ **Answer: C**

3. Which level includes both biotic and abiotic factors?

- A. Population
- B. Community
- C. Ecosystem
- D. Species

☒ **Answer: C**

2. Energy Flow & Trophic Levels

4. Energy flow in an ecosystem is:

- A. Cyclic
- B. One-way
- C. Random
- D. Bidirectional

☒ **Answer: B**

5. The primary source of energy for most ecosystems is:

- A. Wind
- B. Soil
- C. Sun
- D. Water

☒ **Answer: C**

6. Producers are also called:

- A. Heterotrophs
- B. Autotrophs
- C. Decomposers
- D. Consumers

☒ **Answer: B**

7. According to the 10% law, energy transferred to the next trophic level is about:

- A. 50%
- B. 25%
- C. 10%
- D. 1%

☒ **Answer: C**

8. Which organism is a secondary consumer?

- A. Grass
- B. Rabbit
- C. Frog
- D. Hawk

☒ **Answer: C**

3. Population Growth Models

9. Exponential growth occurs when:

- A. Resources are limited
- B. Resources are unlimited
- C. Population declines
- D. Birth rate equals death rate

☒ **Answer: B**

10. Logistic growth includes:

- A. No limits
- B. Carrying capacity
- C. Constant decline
- D. No reproduction

☒ **Answer: B**

11. Carrying capacity refers to:

- A. Maximum population size environment can support
- B. Minimum population size
- C. Average population size
- D. Migration rate

☒ **Answer: A**

12. The shape of exponential growth curve is:

- A. S-shaped
- B. J-shaped
- C. Linear
- D. Circular

☒ **Answer: B**

4. Species Interactions

13. In mutualism:

- A. One benefits, other harmed
- B. Both benefit
- C. Both harmed
- D. One unaffected

☒ **Answer: B**

14. Predator-prey interaction is:

- A. Mutualism
- B. Competition
- C. Predation
- D. Commensalism

☒ **Answer: C**

15. In commensalism:

- A. Both benefit
- B. One benefits, other unaffected
- C. Both harmed
- D. One dies

☒ **Answer: B**

16. Competition occurs when:

- A. Species share same niche
- B. Species avoid each other
- C. Species cooperate
- D. Species mutate

☒ **Answer: A**

5. Community & Ecosystem Concepts

17. A food chain shows:

- A. Energy flow

- B. Genetic variation
- C. Evolution
- D. Mutation

☒ **Answer: A**

18. Food webs are:

- A. Linear
- B. Simple
- C. Interconnected food chains
- D. Closed loops

☒ **Answer: C**

19. Decomposers mainly:

- A. Produce energy
- B. Break down organic matter
- C. Hunt animals
- D. Photosynthesize

☒ **Answer: B**

20. Ecological succession is:

- A. Sudden extinction
- B. Gradual change in species composition
- C. Mutation
- D. Migration

☒ **Answer: B**



UNIT 2: EVOLUTION – MCQs

1. Origins of Life

21. Early Earth atmosphere likely lacked:

- A. Methane
- B. Oxygen
- C. Ammonia
- D. Hydrogen

☒ **Answer: B**

22. The Miller-Urey experiment showed:

- A. Evolution of humans
- B. Organic molecules can form under early Earth conditions
- C. DNA replication

D. Natural selection

☒ **Answer: B**

2. Natural Selection

23. Natural selection was proposed by:

- A. Mendel
- B. Darwin
- C. Watson
- D. Lamarck

☒ **Answer: B**

24. Natural selection acts on:

- A. Individuals
- B. Populations
- C. Phenotypes
- D. Genotypes only

☒ **Answer: C**

25. “Survival of the fittest” means:

- A. Strongest survive
- B. Best adapted survive
- C. Largest survive
- D. Fastest survive

☒ **Answer: B**

26. Variation in population arises from:

- A. Mutation
- B. Recombination
- C. Both A and B
- D. None

☒ **Answer: C**

3. Macroevolution

27. Macroevolution refers to:

- A. Small changes
- B. Large-scale evolutionary changes
- C. Cellular changes

D. Mutation only

☒ **Answer: B**

28. Speciation occurs when:

A. Species interbreed

B. New species form

C. Populations decrease

D. Mutation stops

☒ **Answer: B**

29. Geographic isolation leads to:

A. Extinction

B. Speciation

C. Mutation

D. Competition

☒ **Answer: B**

4. Evidence of Evolution

30. Fossils provide evidence of:

A. Future organisms

B. Past life forms

C. Genetic mutation

D. Photosynthesis

☒ **Answer: B**

31. Homologous structures indicate:

A. Same function

B. Common ancestry

C. Different ancestry

D. Mutation

☒ **Answer: B**

32. Analogous structures show:

A. Same origin

B. Same function, different origin

C. No similarity

D. Evolution stops

☒ **Answer: B**

33. Vestigial organs are:

A. Fully functional

B. Reduced structures with little function

- C. New organs
- D. Harmful organs

☒ **Answer: B**

Mixed Conceptual Questions

34. Which process links ecology and evolution?

- A. Mutation
- B. Natural selection
- C. Photosynthesis
- D. Respiration

☒ **Answer: B**

35. Which trophic level has the most energy?

- A. Top predators
- B. Producers
- C. Secondary consumers
- D. Decomposers

☒ **Answer: B**

36. Adaptation improves:

- A. Survival and reproduction
- B. Mutation rate
- C. Energy loss
- D. Competition only

☒ **Answer: A**

ASSERTION–REASON MCQs **(ECOLOGY + EVOLUTION)**

Format:

- A = Both true, R explains A
 - B = Both true, R does NOT explain A
 - C = A true, R false
 - D = A false, R true
-

Ecology

1.

Assertion: Energy flow in an ecosystem is unidirectional.

Reason: Energy is lost as heat at each trophic level.

☒ Answer: A

2.

Assertion: Food webs are more stable than food chains.

Reason: They provide alternative feeding pathways.

☒ Answer: A

3.

Assertion: Logistic growth shows an S-shaped curve.

Reason: Resources are unlimited in logistic growth.

☒ Answer: C

4.

Assertion: Producers occupy the first trophic level.

Reason: They convert solar energy into chemical energy.

☒ Answer: A

5.

Assertion: Competition reduces population size.

Reason: Organisms compete for limited resources.

☒ Answer: A

Evolution

6.

Assertion: Natural selection leads to adaptation.

Reason: Individuals with favorable traits survive and reproduce.

☒ Answer: A

7.

Assertion: Mutations are always harmful.

Reason: They change DNA sequence.

☒ Answer: D

8.

Assertion: Homologous structures indicate common ancestry.

Reason: They perform similar functions.

☒ **Answer: C**

9.

Assertion: Speciation can occur due to geographic isolation.

Reason: Isolation prevents gene flow.

☒ **Answer: A**

10.

Assertion: Fossils are direct evidence of evolution.

Reason: They show transitional forms.

☒ **Answer: A**



TRICKY / HIGH-LEVEL MCQs



Ecology

11. Which pyramid is always upright?

- A. Pyramid of energy
- B. Pyramid of biomass
- C. Pyramid of numbers
- D. All

☒ **Answer: A**

12. If primary productivity increases, what happens?

- A. Decrease in consumers
- B. Increase in energy available to higher trophic levels
- C. No effect
- D. Ecosystem collapse

☒ **Answer: B**

13. A keystone species is one that:

- A. Is most abundant
- B. Has disproportionate impact on ecosystem
- C. Is smallest
- D. Is producer

☒ **Answer: B**

14. Which interaction benefits one species and harms another?

- A. Mutualism
- B. Parasitism
- C. Commensalism

D. Neutralism

☒ **Answer: B**

Evolution

15. Which is NOT a source of genetic variation?

- A. Mutation
- B. Recombination
- C. Natural selection
- D. Gene flow

☒ **Answer: C**

16. Hardy-Weinberg equilibrium assumes:

- A. Natural selection occurs
- B. No mutation
- C. Small population
- D. Migration

☒ **Answer: B**

17. Analogous structures arise due to:

- A. Divergent evolution
- B. Convergent evolution
- C. Mutation
- D. Isolation

☒ **Answer: B**

18. Adaptive radiation is:

- A. Evolution of one species into many
- B. Extinction
- C. Mutation
- D. Competition

☒ **Answer: A**

SUPER QUICK REVISION (1-PAGE STYLE)

Ecology Cheat Sheet

- **Levels:** Organism → Population → Community → Ecosystem → Biosphere
 - **Energy Flow:** Sun → Producers → Consumers → Decomposers
 - **10% Law:** Only 10% energy transferred
 - **Growth Models:**
 - Exponential → J curve
 - Logistic → S curve (with carrying capacity K)
 - **Interactions:**
 - Mutualism (+,+)
 - Parasitism (+,-)
 - Commensalism (+,0)
 - Competition (-,-)
 - **Key Concepts:**
 - Food chain vs food web
 - Ecological succession
 - Keystone species
-



Evolution Cheat Sheet

- **Origin of Life:** Chemical evolution (Miller-Urey)
- **Natural Selection:** Variation + selection + inheritance
- **Fitness:** Reproductive success
- **Macroevolution:** Speciation, adaptive radiation
- **Evidence:**
 - Fossils
 - Homologous structures
 - Analogous structures
 - Vestigial organs
- **Key Terms:**
 - Mutation = source of variation
 - Gene flow = migration of genes
 - Genetic drift = random change



UNIT 1: ECOLOGY

1. Levels of Ecological Organization

1. Smallest ecological unit → **Organism**
2. Same species group → **Population**
3. Different species together → **Community**
4. Biotic + abiotic → **Ecosystem**

2. Energy Flow & Trophic Levels

- Energy flow → **Unidirectional**
 - Source → **Sun**
 - 10% Law → Only 10% energy transferred
 - Producers → Autotrophs
 - Consumers → Heterotrophs
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3. Population Growth

- Exponential → **J-shaped curve**
 - Logistic → **S-shaped curve**
 - Carrying capacity (K) → Max sustainable population
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4. Species Interactions

Type	Effect
Mutualism	+ +
Parasitism	+ -
Commensalism	+ 0
Competition	- -

5. Ecosystem Concepts

- Food chain → Linear
 - Food web → Complex
 - Decomposers → Nutrient recycling
 - Succession → Gradual ecosystem change
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UNIT 2: EVOLUTION

1. Origin of Life

- Early atmosphere → No oxygen
- Miller-Urey → Organic molecules formed

2. Natural Selection

- Proposed by → **Darwin**
- Acts on → **Phenotype**
- Fitness → Reproductive success

3. Macroevolution

- Speciation → New species formation
- Isolation → Prevents gene flow

4. Evidence

- Fossils → Past life
 - Homologous → Common ancestry
 - Analogous → Same function
 - Vestigial → Reduced function
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MCQs SECTION

Basic MCQs (Sample)

1. Energy flow is:
A. Cyclic B. One-way C. Random D. Reversible
☒ B
2. Carrying capacity means:
A. Minimum size B. Maximum sustainable size
C. Growth rate D. Migration
☒ B
3. Natural selection acts on:
A. DNA B. Phenotype C. Cells D. Organs
☒ B
4. Fossils show:
A. Future life B. Past life C. Mutation D. Energy
☒ B
